**RAJALAKSHMI ENGINEERING COLLEGE**
*Approved by AICTE | Affiliated to Anna University | Accredited by NAAC*

Register No.: _____

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| Continuous Assessment-I |
| Artificial Intelligence and Machine Learning (III sem) |
| Degree: B. E / B. Tech Semester: III &V Year: II &III |
| Course code & Title: AI23231 & Principles of Artificial Intelligence |
| Date: 02.09.26 QP CODE: 151104 |
| [Common to: AIDS (III sem) & IT (V sem)] |

*[Regulations: 2023]*

**Time**: 2 hr 15 min **Max. Marks: 75**

Answer All Questions

**Part A: 10 x 2 = 20 Marks**

| | | | |
|---|---|---|---|
| 1 | What is meant by rationality of an intelligent agent? | [CO1 ] | [K1] |
| 2 | Classify an environment that is fully observable, deterministic and static. | [CO1 ] | [K2] |
| 3 | Compare simple reflex agents and learning agents based on their ability to improve performance. | [CO1 ] | [K2] |
| 4 | A delivery robot selects actions based on both delivery time and energy consumption. Identify the most suitable type of agent. | [CO1 ] | [K3] |
| 5 | What is Depth-Limited Search? | [CO2 ] | [K1] |
| 6 | Why can Bidirectional Search reduce the number of nodes explored compared with a one-directional search? | [CO2 ] | [K2] |
| 7 | A search problem has a very large search space but limited memory. Select an appropriate memory-bounded heuristic search approach. | [CO2 ] | [K3] |
| 8 | Contrast Hill Climbing and Local Beam Search based on the number of states maintained during search. | [CO2 ] | [K2] |
| 9 | Define a game tree in adversarial search. | [CO3 ] | [K1] |
| 10 | Why is a utility function required in game-playing problems? | [CO3 ] | [K2] |

**Part B: 5x11=55 Marks**

| | | | | |
|---|---|---|---|---|
| 11 | a. | Describe the architecture of an intelligent agent. Explain different types of agents with suitable examples. | [CO1 ] | [K2 ] |

**| | K1 – Remember; K2 – Understand; K3 – Apply; K4 – Analyze; K5 – Evaluate; K6 – Create**

**[OR]**

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|---|---|---|---|---|---|
| | b. | Explain the structure of a learning agent and discuss the functions of its learning element, performance element, critic and problem generator. | [CO1 ] | [K2 ] |

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| 12 | a. | Compare goal-based and utility-based agents with respect to action selection, decision quality and suitability for uncertain environments. | [CO1 ] | [K4 ] |

**[OR]**

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|---|---|---|---|---|---|
| | b. | A robot must travel from S to G through the following paths:<br>S → A → C → G<br>S → A → D → G<br>S → B → E → G<br>The corresponding path costs are 9, 12 and 7 respectively.<br>Formulate the task as a state-space problem and determine the preferred solution when path cost is used as the performance criterion. | [CO1 ] | [K4 ] |

| | | | | | |
|---|---|---|---|---|---|
| 13 | a. | Explain Depth-Limited Search and Bidirectional Search. State their working principles, advantages and limitations. | [CO2 ] | [K2 ] |

**[OR]**

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|---|---|---|---|---|---|
| | b. | Compare *A Search and AO* Search** based on the type of graph handled, solution representation and node-selection strategy. | [CO2 ] | [K2 ] |

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|---|---|---|---|---|---|
| 14 | a. | Consider the following heuristic values:<br>The connections are:<br>S → A, B<br>A → C, D<br>B → D | [CO2 ] | [K3 ] |

| Node | S | A | B | C | D | G |
|---|---|---|---|---|---|---|
| h(n) | 8 | 5 | 6 | 2 | 4 | 0 |

$C \rightarrow G$

$D \rightarrow G$

Apply Greedy Best-First Search and show the order in which nodes are selected until the goal is reached.

**[OR]**

b. Explain Local Beam Search and discuss how maintaining multiple candidate states distinguishes it from Hill Climbing Search. [CO2 ] [K2 ]

15 a. Consider a CSP with variables A, B and C, each having the domain {Red, Green, Blue}, subject to: [CO3 ] [K3]

- $A \neq B$
- $B \neq C$
- $A \neq C$

Determine all valid assignments when A = Red.

**[OR]**

b. Consider a game tree in which a MAX node has two MIN children: [CO3 ] [K3 ]

- First MIN node has terminal values 8, 3, 5
- Second MIN node has terminal values 6, 4, 7

Determine the Minimax value of the root. Then analyse which branches can be avoided if Alpha-Beta pruning is applied using an appropriate left-to-right evaluation order.